

- RAG Learning

1.2 The Market

Business — Edexcel (9BS0) — Theme 1: Marketing and people

This topic gives you the economic tools that sit underneath every marketing decision: what determines demand and supply, how the two interact to set a price, and how sensitive demand is to changes in price and income. It is the most quantitative part of Theme 1 — elasticity calculations appear regularly as 4-mark “calculate” questions on Paper 1 — and elasticity is one of the most useful concepts for evaluation anywhere in the course, because it tells you whether a price change will raise or lower revenue.

Demand — the quantity of a good or service that consumers are willing and able to buy at a given price in a given period.

Supply — the quantity that producers are willing and able to offer for sale at a given price in a given period.

Equilibrium price — the price at which quantity demanded equals quantity supplied, so the market clears.

Substitute — a good that can be used in place of another (Coca-Cola and Pepsi).

Complement — a good used together with another (games consoles and games).

Price elasticity of demand (PED) — the responsiveness of quantity demanded to a change in price.

Income elasticity of demand (YED) — the responsiveness of quantity demanded to a change in real income.

1.2.1 Demand

A change in the **price** of the product causes a **movement along** the demand curve (an extension or contraction of demand). A change in any **other** factor causes a **shift** of the whole curve: to the right if demand increases at every price, to the left if it falls.

Factor	Effect on demand	Example
Price of substitutes	If a substitute's price rises, demand for this product increases (shift right)	Rising rail fares increase demand for coach travel
Price of complements	If a complement's price rises, demand for this product falls (shift left)	Higher petrol prices reduce demand for large petrol cars
Consumer incomes	Higher real incomes raise demand for normal goods; lower demand for inferior goods	Falling real incomes in 2022–23 boosted discounters and own-label ranges
Fashions, tastes and preferences	Products that become fashionable see demand rise	Growth of low- and no-alcohol beer; decline of demand for DVDs
Advertising and branding	Successful campaigns shift demand right and make it less price-sensitive	John Lewis Christmas adverts
Demographics	Changes in the size, age and make-up of the population	Ageing population raises demand for care homes and mobility aids
External shocks	Sudden unforeseen events outside the firm's control	Covid-19 lockdowns: demand for home-fitness equipment soared, demand for air travel collapsed
Seasonality	Predictable variation over the year	Ice cream in summer, toys before Christmas, holidays in school breaks

1.2.2 Supply

A change in the product's own price causes a movement along the supply curve. Other factors shift the curve: a **rightward** shift means more is supplied at every price (supply increases); a

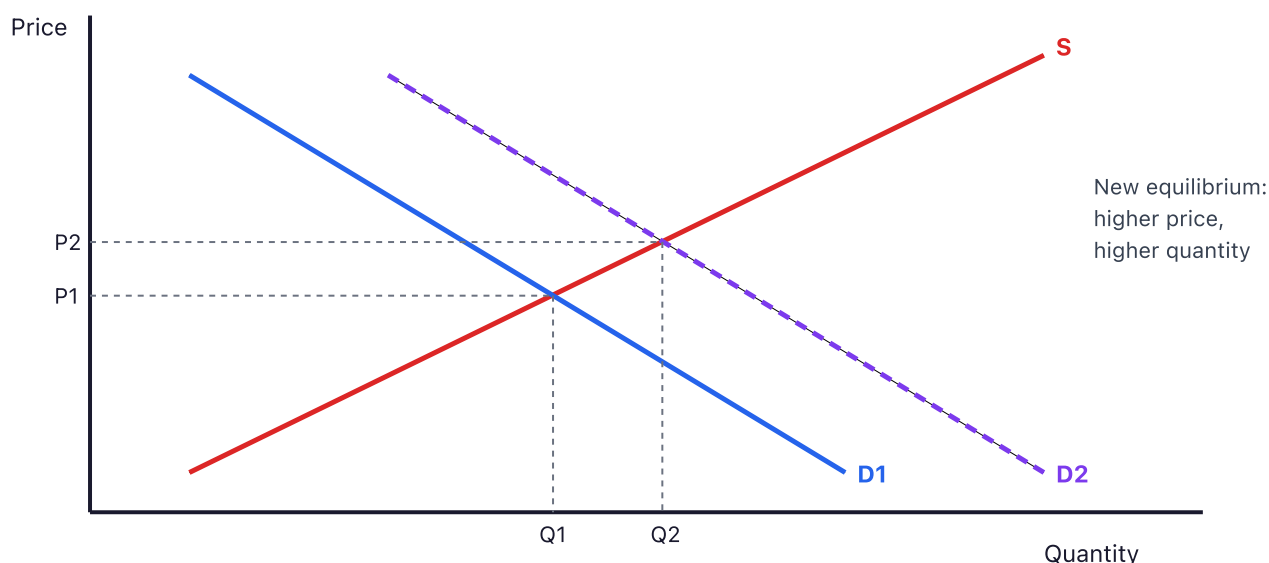
leftward shift means supply falls.

Factor	Effect on supply	Example
Costs of production	Higher costs (wages, materials, energy) reduce supply (shift left)	2022 energy price rises cut output at energy-intensive firms such as bakeries and glassmakers
New technology	Lowers unit costs and raises output: supply shifts right	Automation in warehouses; falling cost of solar panels
Indirect taxes	Taxes on spending (VAT, excise duty) raise costs to suppliers: shift left	Soft Drinks Industry Levy (2018) — many producers reformulated to avoid it
Government subsidies	Payments that reduce producers' costs: shift right	Subsidies for renewable energy generation; heat-pump grants
External shocks	Unforeseen events that disrupt production	Poor harvests raising cocoa prices in 2024; Suez Canal blockage 2021

1.2.3 Markets — the interaction of supply and demand

The equilibrium price is where the demand and supply curves cross. If the price is set **above** equilibrium there is **excess supply** (a surplus), so firms cut prices to clear stock. If it is **below** equilibrium there is **excess demand** (a shortage), so prices are bid up — think of concert tickets resold at far above face value.

An increase in demand: D1 shifts to D2



b) Drawing and interpreting supply and demand diagrams

Change	Shift	Effect on price	Effect on quantity
Increase in demand	D shifts right	Rises	Rises
Decrease in demand	D shifts left	Falls	Falls
Increase in supply	S shifts right	Falls	Rises
Decrease in supply	S shifts left	Rises	Falls

Worked interpretation: poor West African harvests in 2023–24 cut the supply of cocoa (S shifts left). With demand broadly unchanged, the world cocoa price more than doubled and quantity traded fell. For a chocolate maker such as Mondelez or Hotel Chocolat this is a cost increase: it must choose between raising prices, shrinking pack sizes (“shrinkflation”) or accepting lower margins — and the right choice depends on the PED of its products.

Exam Tip: When drawing a diagram: label the axes (Price, Quantity), label each curve, show the original and new equilibrium with dotted lines to both axes, and use arrows to show the direction of the shift. Then explain the diagram in words and link it to the business in the question — the diagram alone does not earn analysis marks.

1.2.4 Price elasticity of demand (PED)

a) Calculation

$$\text{PED} = \% \text{ change in quantity demanded} \div \% \text{ change in price}$$

$$\% \text{ change} = (\text{new} - \text{old}) \div \text{old} \times 100$$

Worked example. A gym raises its monthly membership from £40 to £44. Members fall from 2,000 to 1,760.

- % change in price = $(44 - 40) \div 40 \times 100 = +10\%$
- % change in quantity = $(1,760 - 2,000) \div 2,000 \times 100 = -12\%$
- PED = $-12 \div +10 = -1.2$ (price elastic)

Using PED to predict: if PED = -0.4 and the price rises by 5%, quantity demanded changes by $-0.4 \times 5 = -2\%$. If current sales are 50,000 units, new sales = $50,000 \times 0.98 = 49,000$.

b) Interpretation of numerical values

Value (ignoring the minus sign)	Description	Meaning
Greater than 1	Price elastic	Quantity changes by a larger percentage than price
Less than 1 (between 0 and 1)	Price inelastic	Quantity changes by a smaller percentage than price
Exactly 1	Unitary elasticity	Quantity changes by the same percentage as price; revenue unchanged
0	Perfectly inelastic	Quantity does not change at all (theoretical)
Infinity	Perfectly elastic	Any price rise loses all sales (theoretical)

PED is normally **negative** because price and quantity demanded move in opposite directions. Edexcel mark schemes accept the value with or without the sign if it is interpreted correctly, but stating the sign is good practice.

c) Factors influencing PED

- **Number and closeness of substitutes** — the most important factor. One brand of petrol is price elastic; petrol as a whole is inelastic.
- **Brand loyalty and differentiation** — strong brands (Apple, Heinz) have more inelastic demand.
- **Necessity versus luxury** — necessities are inelastic.
- **Proportion of income spent** — a 10% rise on a box of matches goes unnoticed; on a car it does not.
- **Habit and addiction** — cigarettes, coffee.
- **Time** — demand becomes more elastic over time as consumers find alternatives.
- **Who pays** — business travellers whose employer pays are less price-sensitive.

d) Significance for pricing decisions

- If demand is **inelastic**, a firm can **raise** prices to increase revenue (and, as fewer units are made, costs fall too, so profit rises strongly).
- If demand is **elastic**, a **price cut** increases revenue — but profit rises only if the extra contribution exceeds any extra costs of producing more.
- Firms try to make demand more inelastic through branding, differentiation and loyalty schemes.
- PED helps a firm decide how much of a cost increase it can pass on to customers.

e) PED and total revenue

Total revenue = Price × Quantity sold

	Price rise	Price cut
Elastic (>1)	Revenue falls	Revenue rises

Inelastic (<1)	Revenue rises	Revenue falls
Unitary (=1)	No change	No change

Worked example (gym above): revenue before = $£40 \times 2,000 = £80,000$ per month. After = $£44 \times 1,760 = £77,440$. Revenue fell by $£2,560$ because demand is elastic.

Second example: a train operator on a commuter route with $PED = -0.3$ raises fares 6% from $£10$. Passengers fall 1.8% from 10,000 to 9,820. Revenue rises from $£100,000$ to $£10.60 \times 9,820 = £104,092$ — commuters have few substitutes.

Exam Tip: A common 4-mark question asks for the change in revenue after a price change given PED. Show every step: % change in quantity ($PED \times \% \text{ price change}$), new quantity, new price, new revenue, then the difference. Method marks are awarded even if the final answer is wrong, so never write just a number.

Evaluating PED: estimates are based on past data and assume other factors stay constant (*ceteris paribus*), which rarely happens; PED differs at different prices; competitors may respond to a price change; and it is costly to measure accurately. It is a guide, not a guarantee.

1.2.5 Income elasticity of demand (YED)

a) Calculation

$$YED = \% \text{ change in quantity demanded} \div \% \text{ change in real income}$$

Worked example. Real household incomes rise by 4%. Demand for foreign holidays rises by 10%; demand for bread rises by 1%; demand for supermarket value-range products falls by 2%.

- Holidays: $10 \div 4 = +2.5$ (luxury / income elastic)
- Bread: $1 \div 4 = +0.25$ (normal, income inelastic necessity)
- Value range: $-2 \div 4 = -0.5$ (inferior good)

b) Interpretation

YED value	Type of good	Behaviour as income rises
Greater than +1	Normal, luxury (income elastic)	Demand rises more than proportionately
Between 0 and +1	Normal, necessity (income inelastic)	Demand rises less than proportionately
Negative	Inferior good	Demand falls; consumers trade up

Note the **sign matters** for YED (unlike PED): it tells you whether the good is normal or inferior.

c) Factors influencing YED

- Whether the product is a necessity or a luxury
- Availability of higher-quality alternatives that people trade up to (making the cheaper good inferior)
- Consumers' current level of income — a car may be a luxury in a low-income economy and a necessity in a high-income one
- Proportion of income spent on the product

d) Significance to businesses

- **Sales forecasting** — firms can predict how sales will respond to the business cycle (see 2.5). In a recession, sellers of luxuries (fine dining, cruises) suffer most; discounters (Aldi, Poundland historically) may gain.
- **Product portfolio** — holding a mix of income-elastic and inferior products spreads risk across the cycle (Tesco's Finest and value ranges).
- **Market entry** — fast-growing economies with rising incomes offer strong demand for income-elastic goods (links to 4.1).
- **Stock and staffing decisions** — planning capacity ahead of expected changes in incomes.

Exam Tip: Evaluate YED by noting that incomes are only one influence on demand: in 2022–23 real incomes fell, yet demand for some luxuries held up among higher earners with

savings. YED is an average across all consumers and is based on past behaviour.

Common mistakes

- Saying a price change "shifts the demand curve" — it causes a movement along it.
- Dividing the wrong way round (price over quantity). Quantity is always on top.
- Calling PED of -0.6 "elastic" because it is negative — look at the size, not the sign.
- Assuming higher revenue means higher profit; costs must be considered.
- Ignoring the sign on YED — a negative YED is the whole point (inferior good).
- Forgetting competitor reaction: PED assumes rivals keep their prices constant.

Quick recall

- Demand shifters: substitutes, complements, incomes, tastes, advertising/branding, demographics, external shocks, seasonality
- Supply shifters: costs, technology, indirect taxes, subsidies, external shocks
- Own price → movement along; anything else → shift
- $PED = \% \Delta Q_d \div \% \Delta P$; >1 elastic, <1 inelastic
- Inelastic: raise price, revenue rises. Elastic: cut price, revenue rises
- Main PED factor: number and closeness of substitutes (branding reduces PED)
- $YED = \% \Delta Q_d \div \% \Delta Y$; >1 luxury, $0-1$ necessity, <0 inferior
- Both elasticities: based on past data, ceteris paribus, a guide not a guarantee